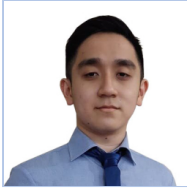


NAUVAL ZULFIKAR



Applied ML / NLP Engineer · Responsible AI candidate Bandung, Indonesia (relocatable to Gothenburg, 2026-09-01)

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[LinkedIn](#) · [GitHub](#) · [Portfolio](#)

Visa status: UK Graduate Visa (active). Indonesian passport. Eligible for EU Researcher (Hosting Agreement) visa for European PhDs.

PROFILE

MSc Business Analytics graduate from Aston University with a First Class (1:1) distinction, specialising in the integration of blockchain and LLMs for supply chain systems. Extensive experience in agentic AI systems, conceptual modelling, and software engineering for AI, aligning with Assoc. Prof. Jennifer Horkoff's research on intentional monitoring of agentic AI software.

EDUCATION

Aston University, Birmingham, UK

MSc Business Analytics, 1:1 First Class · EQF Level 7 · Sep 2023 – Sep 2024

- **1st Winner, Aston Enterprise Scholarship 2023** (full-tuition competitive award)
- **Aston Pitching Contest 2024** runner-up (AI-powered supply chain proposal)
- **Distinctions:** Decision Models, Effective Management Consultancy, Software Analytics
- **Dissertation:** *Enhancing Supply Chain Information Sharing Systems Through Blockchain: Integrated Customer Reviews LLM Analysis: A Smart Contract Approach* (supervised by **Dr. Ammar Al-Bazi**)

Ritsumeikan Asia Pacific University, Beppu, Japan

BBA Marketing · EQF Level 6 · Oct 2016 – Sep 2019

- JASSO Scholarship + APU 50% Tuition Reduction Scholarship
- **Distinctions:** Business Data Analysis, Consumer Behaviour, Marketing Research

RESEARCH-RELEVANT PROJECTS

Mini Digital Twin for Integrated Passenger-Freight Transport

- Developed a three-layer Digital Twin prototype using FastAPI, mesa, and simpy, featuring live dashboards and interactive disruption injection. Achieved 45% acceptance and 62% terminal utilisation.
- [GitHub](#)

LLM-Generated Adaptive Shipper Decision Rules

- Created an end-to-end RAG system using gpt-4o-mini, integrating literature-grounded decision rules for crowd-shipping.
- [GitHub](#)

Crowd-Shipping ABM Behavioural Sweep

- Conducted a 450-run parameter sweep using mesa to evaluate behavioural modelling in crowd-shipping scenarios.
- [GitHub](#)

VRPOD MILP Optimisation

- Implemented MILP optimisation for VRPOD, achieving a 38.5% cost reduction on a Birmingham-inspired instance.
- [GitHub](#)

INDUSTRY EXPERIENCE

Public Works & Spatial Planning Department (PUTR), Bandung

Data Scientist & Project Manager · Sep 2022 – Present

- Automated ETL pipelines and developed self-service dashboards, improving project delivery by 45% and data efficiency by 20%.

Syncwell, Birmingham, UK

Marketing Data Scientist · Jul – Sep 2024

- Implemented predictive ML and A/B testing, increasing CTR by 40% and follower growth by 22%.

PCOS Challenge, Birmingham, UK

Strategic Data Scientist Specialist · Jun – Aug 2024

- Developed a customer attribution model, reducing manual workload by 80%.

Bank Muamalat Indonesia, Jakarta

Lead Data Scientist · Nov 2019 – Aug 2022

- Developed NLP models for credit analysis, achieving a 5.3% investment return and maintaining model reliability.

PUBLICATIONS

Zulfikar, N. (2025) *Performance Analysis in Sport Footwear Sales Prediction Using Machine Learning*. Working paper comparing ML regressors for retail sales forecasting.

TECHNICAL SKILLS

ML/NLP: PyTorch, HuggingFace Transformers, DeBERTa-v3, gpt-4o-mini, sentence-transformers, classification, regression, time-series forecasting.

Languages/Infrastructure: Python, SQL, R, PySpark, NumPy, Scipy, pandas, scikit-learn, FastAPI, Streamlit.

Mathematical/Formal: MILP, Agent-Based Modelling, Discrete-Event Simulation, Digital Twin architecture, Bayesian methods.

Governance: Model validation, reliability, audit documentation, AWS, Docker, Git, ETL/orchestration.

LANGUAGES

Indonesian: Native

English: Fluent (IELTS 7.0 overall)

Japanese: Intermediate

REFEREES

Dr. Ammar Al-Bazi

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Dr. Viktor Pekar

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